

Macroeconomic Shifts and GDP Dynamics in Emerging & Regional Markets (2015–2025)

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Dataset: World Bank Open Data API (NY.GDP.MKTP.KD.ZG)
Pipeline: Python, Pandas, Plotly (Hosted on GitHub)

Executive Summary
This research brief evaluates annual Real GDP growth rates across primary monitored economies—including Iraq, Saudi Arabia, Türkiye, Germany, and the Russian Federation—from 2015 to 2025. By executing a custom Python data pipeline accessing World Bank API endpoints, this study isolates external macroeconomic shocks, commodity price volatility, and national structural transitions using empirical historical metrics.

Empirical Findings & Data Analysis

1. Hydrocarbon Vulnerability & Macro Shocks (Iraq & Saudi Arabia)

As visualized in Figure 1, oil-exporting economies experienced sharp contraction troughs during the 2020 global energy demand slump:

- Iraq: Plunged to a severe trough of -11.3% in 2020 due to oil price crashes, followed by a post-pandemic recovery reaching 7.0% in 2022 before stabilizing near historical baseline levels.
- Saudi Arabia: Mirrored this volatility, contracting by -4.1% in 2020 before expanding to 8.7% in 2022, supported by non-oil revenue growth and energy market recalibrations.

2. Industrial Drag vs. Domestic Expansion (Germany & Türkiye)

- Germany: Exhibited sustained industrial friction, contracting to -3.8% in 2020 and remaining constrained near stagnant levels (-0.3% in 2023) due to elevated input energy costs.
- Türkiye: Maintained aggressive domestic growth momentum, remaining positive in 2020 (1.9%) and surging to a peak of 11.4% in 2021—the highest point on the multi-year index.

3. Sanctions & Trade Realignment (Russian Federation)

- Russian Federation: Contracted to -2.7% in 2022 following international trade restrictions, followed by a stabilization trend reaching 3.6% in 2023 as trade corridors realigned toward Asian markets.

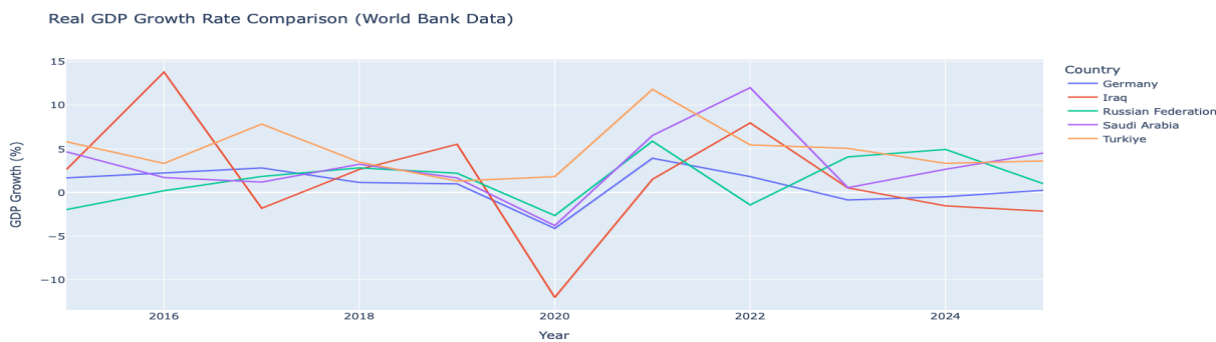


Figure 1: Comparative Real GDP Growth Dynamics (2015–2025) generated via Plotly using direct World Bank API metrics.

Technical Pipeline Implementation

The quantitative foundation of this research relies on a three-tier automated data architecture:

1. API Ingestion: Direct querying of World Bank REST API endpoints using indicator code NY.GDP.MKTP.KD.ZG.
2. Data Wrangling: Automated cleaning, missing-value handling, and time-series normalization using Python (Pandas).
3. Visual Analytics: Rendering dynamic, multi-country trend models using Plotly to visually capture regional growth divergences.

Conclusion & Policy Implications

Empirical findings confirm that mono-resource markets remain hyper-vulnerable to exogenous price shocks, whereas economies with diversified manufacturing bases exhibit faster recovery trajectories. Long-term macroeconomic resilience demands targeted non-resource diversification and continuous API-driven monitoring systems.